

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built **backend automation** for compliance-sensitive MCA filing workflows with browser execution, public-site API lookups, retries, **session recovery**, validation checks, and operator checkpoints, reducing manual filing time by **85%**.
- Implemented an **AI-assisted draft-generation flow** for company-name suggestions, object descriptions, and filing-field drafts, keeping generated outputs inspectable before deterministic workflow execution.
- Modeled screenshots, generated drafts, API responses, form state, and failure checkpoints as recoverable workflow records for debugging unreliable external systems.
- Shipped Dockerized Node.js/Go **backend services** on **Google Cloud Run** with **GitHub Actions** release workflows, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Integrated **model-inference workflows** for **95 anatomical brain regions**, wiring PyTorch/TorchScript, ONNX/TinyGrad execution paths, preprocessing, generated artifacts, progress reporting, and correctness checks into Invesalio.
- Reduced runtime by **87%** through async/parallel execution while debugging orientation, label correctness, GPU/CPU paths, large artifacts, and reproducible output inspection.

Open Source

Contributions

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway**, **Kyverno**, **Invesalio**, **Tiled**, **CRIU**, and **VideoLAN**, spanning backend failover behavior, parser robustness, security validation, compatibility fixes, and data/artifact correctness.

Projects

Careers.pointblank.club | Next.js, TypeScript, Firebase, Search, AI-assisted Resume Analysis

- Built a **student-developer recruiting directory** for Point Blank with profile discovery, domain-based search, candidate/project presentation, and recruiter-facing navigation for finding relevant student developers.
- Worked on **AI-assisted resume analysis** to extract skills, project signals, role fit, and profile summaries into a structured recruiting surface instead of leaving matching as raw resume reading.

Hyoka | Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, Next.js, Docker

GitHub

- Built an **AI evaluation/replay control plane** with trace ingestion, validation runs, replay workers, release gates, promotions, project-scoped API keys, audit logs, and a Next.js dashboard.
- Designed **PostgreSQL metadata models** for traces, suites, candidates, gate decisions, artifacts, manifests, and worker leases; added an OpenAI-compatible proxy and OTLP/OpenInference-style ingestion.

Postificus | Go, Echo, React, Vite, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus

GitHub

- Built a **workflow backend** with REST APIs, auth/settings endpoints, PostgreSQL persistence, Redis state/autosave, RabbitMQ workers, S3-compatible storage, and live job/status tracking.
- Added **retries**, **DLQs**, **circuit breakers**, **health checks**, and **Prometheus metrics** so long-running API/browser workflows stayed observable and recoverable under unreliable third-party integrations.

Seaweed Contest Platform | Go, React, Vite, PostgreSQL, Redis, Judge0, Docker, Kubernetes

GitHub

- Built a **backend-heavy platform** for **500+ concurrent users**, covering registration, problem pages, submissions, async judging, admin review, and live ranked leaderboards.
- Integrated Judge0 with async Go dispatch, PostgreSQL result schemas, Redis runtime state, Docker/Kubernetes deployment, and sub-5s **P95** verdict latency.

Technical Skills

Languages: Python, Go, TypeScript, JavaScript

Frameworks: FastAPI, Node.js, Next.js, React, Echo, SQLAlchemy, PyTorch

Tools: PostgreSQL, Redis, RabbitMQ, Docker, GCP Cloud Run, GitHub Actions, OpenTelemetry, Prometheus